

William Francis Arnold

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Education

Korea Advanced Institute of Science and Technology (KAIST)

September 2023 – August 2025

MS, School of AI, MLILAB under Professor Eunho Yang
Research on Transformer Inference & Training Efficiency

University of California, Berkeley

August 2017 – May 2021

BS, Electrical Engineering and Computer Science Major, Math Minor
Graduate Coursework in High Performance Computing, Theoretical Physics

Highest Honors

University Medal Nominee

Projects

Sequence Level Experts for On-Device Infilling - Master's Thesis

Developed and trained hardware-aware MoE modifications for on-device inference by routing sequences rather than tokens. Allows expert offloading to flash, reducing memory usage by 6x while maintaining dense model inference throughput and TTFT latency. Demonstrated model performance through robust training experiments on TPUs using JAX and a custom data pipeline.

WRIST - Torch Auto-Parallelism Search through Simulation

Used torch.fx.Graph traces, network topology analysis, and accelerator performance models to efficiently search millions of parallelism configurations for optimal hardware utilization. Achieved 0.96 spearman correlation predicting 20B transformer training.

Experience

Nvidia – Senior Deep Learning Algorithms Engineer

August 2025 – Current

- Developed novel inference strategies influencing next-gen chip memory and design
- Built multi-region RL functionality in Dynamo for Cognition's RL systems used to train SWE-1.7
- Secured \$80M+ in customer GPU contracts by building inference recipes beating competition by 60%

Google – Student Researcher, Gemini Applied Research

April 2025 – July 2025

- Developed methods for efficient sampling of math/coding LLM benchmarks

Eluvio – Software Engineer

Summer 2020, October 2021 – November 2022

- Built a distributed key management system in Intel SGX/AMD MET securing \$100M+ worth of Fortune 500 media assets.

UC Berkeley – Machine Learning Researcher & Instructor

August 2018 – May 2021

- Conducted research in Prof. Spanos lab on Reinforcement Learning for intelligent control of energy systems in large buildings, leading to a 3 week energy usage experiment at the National University of Singapore.
- Taught discrete math and probability theory (CS70), receiving 4.97/5.0 on student evaluations (department mean=4.4/5).

Procore Technologies – Machine Learning Intern

June 2019 – August 2019

- Developed systems for and built domain-specific image datasets, yielding 40%+ accuracy improvements on edge classes.

Activate Inc., at Google – Technical Lead

February 2016 – August 2018

- Used machine learning to identify high-demand concepts around 50+ locations in San Francisco for a Fortune 500 company.
- Executed \$250k contract building models to predict how Google employees choose on-campus cafeterias to reduce food waste.

Gradescope – Android Developer

September 2014 – July 2016

Publications

- Factored Agents: Decoupling In-Context Learning and Memorization for Robust Tool Use. [Arxiv](https://arxiv.org/abs/2406.14700).
- Chatbot Arena Estimate: towards a generalized performance benchmark for LLM capabilities. In *NAACL 2025*.
- DisruptionBench: Two Advancements in Machine Learning Driven Disruption Prediction. In *Journal of Fusion Energy 2025*.
- Back-to-Basics Revisited: Benchmarking an Expanded Set of RLHF Algorithms. In *NeurIPS SFLLM 2024*.
- Continuous Convolutional Neural Networks for Disruption Prediction in Nuclear Fusion Plasmas. In *NeurIPS CCAI 2023* (1st author).
- Adapting Surprise Minimizing Reinforcement Learning Techniques for Transactive Control. In *Proceedings of the Twelfth ACM International Conference on Future Energy Systems 2021* (1st author).
- Pricing in Prosumer Aggregations using Reinforcement Learning. In *Proceedings of the Twelfth ACM International Conference on Future Energy Systems 2021*.

Activities

Cal Sailing Team – Varsity Skipper and Fleetmaster

August 2018 – May 2021